Kubernetes Task-2

Task Description:

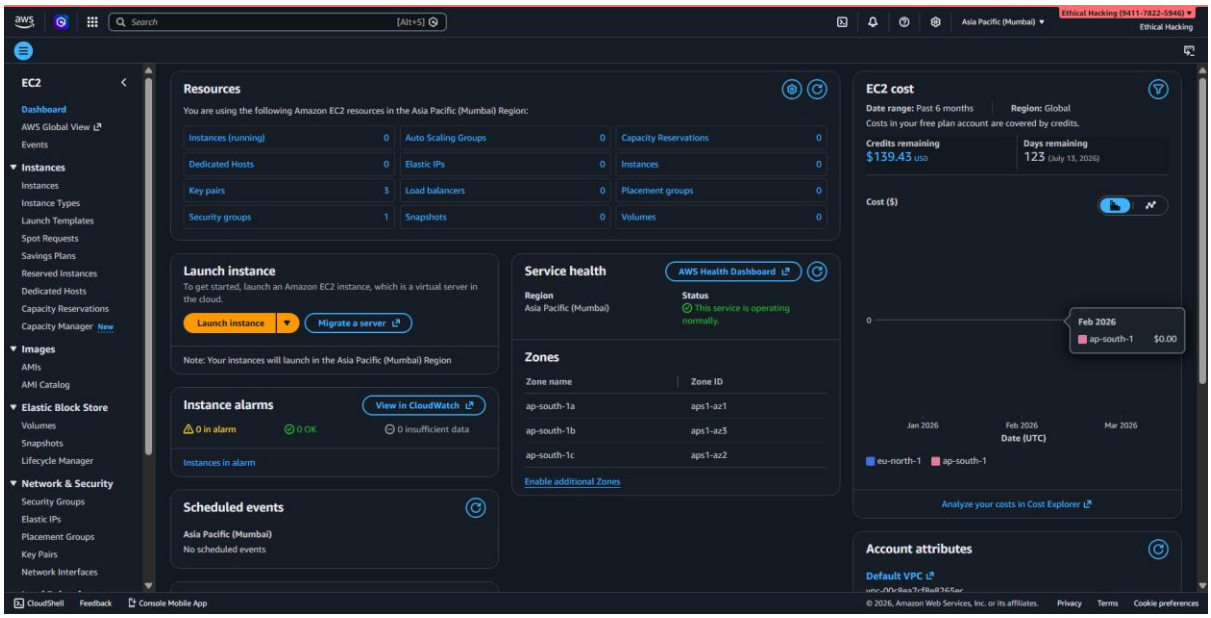
Create the K8s EKS,further you have to do the deployment of the Nginx application and access the application outside the cluster.

Solution:

Creating an K8s cluster using EKS:

```
GHOST@GHOST:~/scriptsAWS/AWS_Scripts$ aws --version
aws-cli/2.34.2 Python/3.13.11 Linux/6.6.87.2-microsoft-standard-WSL2 exe/x86_64.ubuntu.24
kGHOST@GHOST:~/scriptsAWS/AWS_Scripts$ kubectl version
Client Version: v1.35.2
Kustomize Version: v5.7.1
Server Version: v1.35.1
GHOST@GHOST:~/scriptsAWS/AWS_Scripts$ minikube version
minikube version: v1.38.1
commit: c93a4cb9311efc66b90d33ea03f75f2c4120e9b0
GHOST@GHOST:~/scriptsAWS/AWS_Scripts$ eksctl version
0.224.0
GHOST@GHOST:~/scriptsAWS/AWS_Scripts$
```

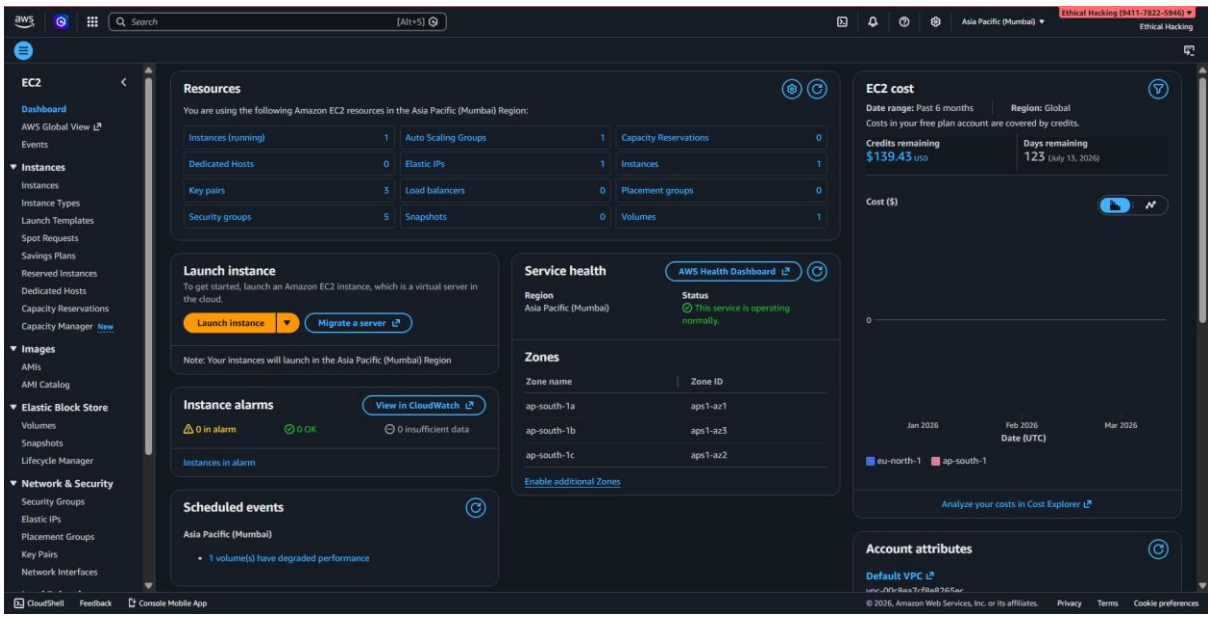
Before creating the cluster and nodes:



Creating an EKS cluster:

```
GHOST@GHOST:~/scriptsAWS/AWS_Scripts$ eksctl create cluster \
--name nginx-cluster \
--region ap-south-1 \
--nodegroup-name nginx-nodes \
--node-type t3.small \
--nodes 1 \
--node-volume-size 20 \
--managed
2026-03-12 16:26:05 [ℹ]  eksctl version 0.224.0
2026-03-12 16:26:05 [ℹ]  using region ap-south-1
2026-03-12 16:26:07 [ℹ]  setting availability zones to [ap-south-1b ap-south-1a ap-south-1c]
2026-03-12 16:26:07 [ℹ]  subnets for ap-south-1b - public:192.168.0.0/19 private:192.168.96.0/19
2026-03-12 16:26:07 [ℹ]  subnets for ap-south-1a - public:192.168.32.0/19 private:192.168.128.0/19
2026-03-12 16:26:07 [ℹ]  subnets for ap-south-1c - public:192.168.64.0/19 private:192.168.160.0/19
2026-03-12 16:26:08 [ℹ]  nodegroup "nginx-nodes" will use "" [AmazonLinux2023/1.34]
2026-03-12 16:26:08 [!]  Auto Mode will be enabled by default in an upcoming release of eksctl. This means managed node groups and managed networking add-ons will no longer be created by default. To maintain current behavior, explicitl
y set 'autoModeConfig.enabled: false' in your cluster configuration. Learn more: https://eksctl.io/usage/auto-mode/
2026-03-12 16:26:08 [ℹ]  using Kubernetes version 1.34
2026-03-12 16:26:08 [ℹ]  creating EKS cluster "nginx-cluster" in "ap-south-1" region with managed nodes
2026-03-12 16:26:08 [ℹ]  will create 2 separate CloudFormation stacks for cluster itself and the initial managed nodegroup
2026-03-12 16:26:08 [ℹ]  if you encounter any issues, check CloudFormation console or try 'eksctl utils describe-stacks --region=ap-south-1 --cluster=nginx-cluster'
2026-03-12 16:26:08 [ℹ]  Kubernetes API endpoint access will use default of {publicAccess=true, privateAccess=false} for cluster "nginx-cluster" in "ap-south-1"
2026-03-12 16:26:08 [ℹ]  CloudWatch logging will not be enabled for cluster "nginx-cluster" in "ap-south-1"
2026-03-12 16:26:08 [ℹ]  you can enable it with 'eksctl utils update-cluster-logging --enable-types={SPECIFY-YOUR-LOG-TYPES-HERE (e.g. all)} --region=ap-south-1 --cluster=nginx-cluster'
2026-03-12 16:26:08 [ℹ]  default addons metrics-server, vpc-cni, kube-proxy, coredns were not specified, will install them as EKS addons
2026-03-12 16:26:08 [ℹ]
2 sequential tasks: { create cluster control plane "nginx-cluster",
    2 sequential sub-tasks: {
        2 sequential sub-tasks: {
            1 task: { create addons },
            wait for control plane to become ready,
        },
        create managed nodegroup "nginx-nodes",
    }
}
2026-03-12 16:26:08 [ℹ]  building cluster stack "eksctl-nginx-cluster-cluster"
2026-03-12 16:26:09 [ℹ]  deploying stack "eksctl-nginx-cluster-cluster"
2026-03-12 16:26:41 [ℹ]  waiting for CloudFormation stack "eksctl-nginx-cluster-cluster"
2026-03-12 16:27:12 [ℹ]  waiting for CloudFormation stack "eksctl-nginx-cluster-cluster"
2026-03-12 16:28:14 [ℹ]  waiting for CloudFormation stack "eksctl-nginx-cluster-cluster"
2026-03-12 16:29:19 [ℹ]  waiting for CloudFormation stack "eksctl-nginx-cluster-cluster"
2026-03-12 16:30:23 [ℹ]  waiting for CloudFormation stack "eksctl-nginx-cluster-cluster"
2026-03-12 16:31:26 [ℹ]  waiting for CloudFormation stack "eksctl-nginx-cluster-cluster"
2026-03-12 16:32:29 [ℹ]  waiting for CloudFormation stack "eksctl-nginx-cluster-cluster"
2026-03-12 16:33:33 [ℹ]  waiting for CloudFormation stack "eksctl-nginx-cluster-cluster"
2026-03-12 16:34:37 [ℹ]  waiting for CloudFormation stack "eksctl-nginx-cluster-cluster"
2026-03-12 16:34:41 [!]  recommended policies were found for "vpc-cni" addon, but since OIDC is disabled on the cluster, eksctl cannot configure the requested permissions; the recommended way to provide IAM permissions for "vpc-cni" ad
don is via pod identity associations; after addon creation is completed, add all recommended policies to the config file, under 'addon.PodIdentityAssociations', and run 'eksctl update addon'
2026-03-12 16:34:41 [ℹ]  creating addon: vpc-cni
2026-03-12 16:34:42 [ℹ]  successfully created addon: vpc-cni
2026-03-12 16:34:42 [ℹ]  creating addon: kube-proxy
2026-03-12 16:34:43 [ℹ]  successfully created addon: kube-proxy
2026-03-12 16:34:43 [ℹ]  creating addon: coredns
2026-03-12 16:34:43 [ℹ]  successfully created addon: coredns
2026-03-12 16:36:54 [ℹ]  building managed nodegroup stack "eksctl-nginx-cluster-nodegroup-nginx-nodes"
2026-03-12 16:36:56 [ℹ]  deploying stack "eksctl-nginx-cluster-nodegroup-nginx-nodes"
2026-03-12 16:36:56 [ℹ]  waiting for CloudFormation stack "eksctl-nginx-cluster-nodegroup-nginx-nodes"
2026-03-12 16:37:28 [ℹ]  waiting for CloudFormation stack "eksctl-nginx-cluster-nodegroup-nginx-nodes"
2026-03-12 16:38:17 [ℹ]  waiting for CloudFormation stack "eksctl-nginx-cluster-nodegroup-nginx-nodes"
```

After creation of cluster and node:



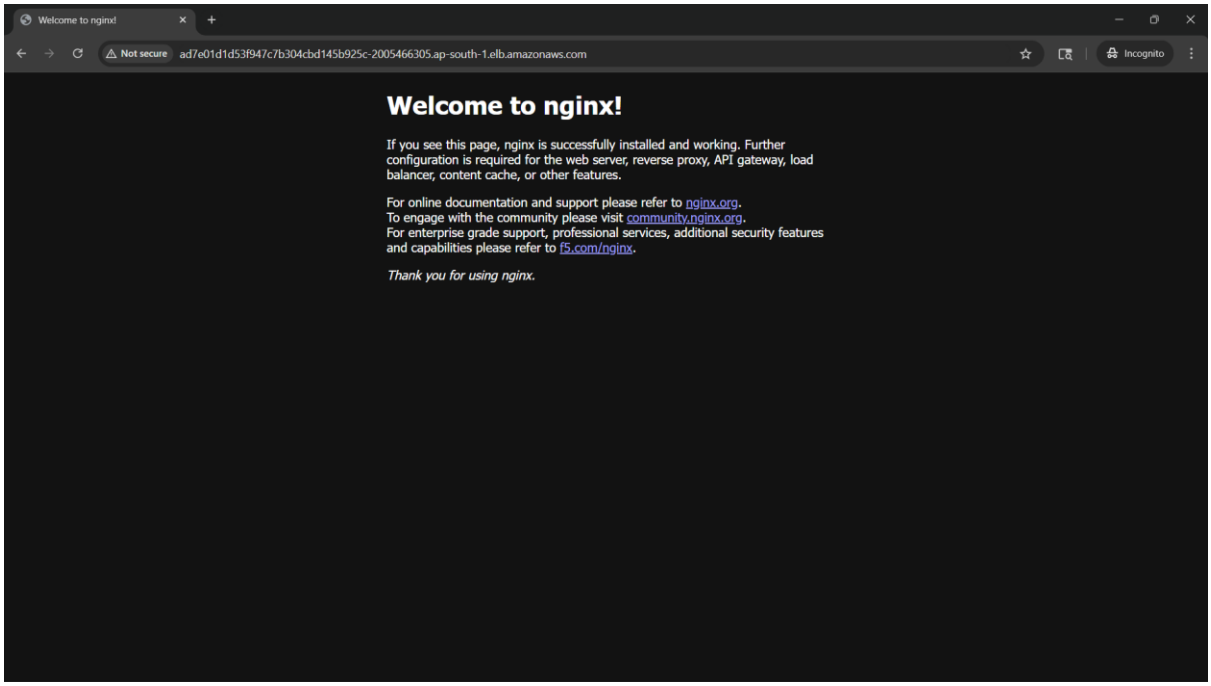

```
GHOST@GHOST: ~/scriptsAW    Ubuntu
GHOST@GHOST:~/scriptsAWS/AWS_Scripts$ kubectl get nodes
NAME                                          STATUS   ROLES    AGE     VERSION
ip-192-168-87-89.ap-south-1.compute.internal  Ready    <none>   2m25s   v1.34.4-eks-f69f56f
GHOST@GHOST:~/scriptsAWS/AWS_Scripts$
```

Deploying an Nginx Application:

```
GHOST@GHOST: ~/scriptsAW   Ubuntu
GHOST@GHOST:~/scriptsAWS/AWS_Scripts$ kubectl create deployment nginx --image=nginx
deployment.apps/nginx created
GHOST@GHOST:~/scriptsAWS/AWS_Scripts$ kubectl get pods
NAME                     READY   STATUS    RESTARTS   AGE
nginx-66686b6766-hkcb5   1/1     Running   0          7s
GHOST@GHOST:~/scriptsAWS/AWS_Scripts$ kubectl expose deployment nginx --type=LoadBalancer --port=80
service/nginx exposed
GHOST@GHOST:~/scriptsAWS/AWS_Scripts$ kubectl get svc
NAME         TYPE           CLUSTER-IP      EXTERNAL-IP                                                              PORT(S)        AGE
kubernetes   ClusterIP      10.100.0.1      <none>                                                                   443/TCP        25m
nginx        LoadBalancer   10.100.217.55   ad7e01d1d53f947c7b304cbd145b925c-2005466305.ap-south-1.elb.amazonaws.com   80:31947/TCP   8s
GHOST@GHOST:~/scriptsAWS/AWS_Scripts$ kubectl get all
NAME                         READY   STATUS    RESTARTS   AGE
pod/nginx-66686b6766-hkcb5   1/1     Running   0          2m8s

NAME                 TYPE           CLUSTER-IP      EXTERNAL-IP                                                              PORT(S)        AGE
service/kubernetes   ClusterIP      10.100.0.1      <none>                                                                   443/TCP        27m
service/nginx        LoadBalancer   10.100.217.55   ad7e01d1d53f947c7b304cbd145b925c-2005466305.ap-south-1.elb.amazonaws.com   80:31947/TCP   103s

NAME                    READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/nginx   1/1     1            1           2m9s

NAME                               DESIRED   CURRENT   READY   AGE
replicaset.apps/nginx-66686b6766   1         1         1       2m9s
GHOST@GHOST:~/scriptsAWS/AWS_Scripts$ kubectl describe svc nginx
Name:                     nginx
Namespace:                default
Labels:                   app=nginx
Annotations:              <none>
Selector:                 app=nginx
Type:                     LoadBalancer
IP Family Policy:         SingleStack
IP Families:              IPv4
IP:                       10.100.217.55
IPs:                      10.100.217.55
LoadBalancer Ingress:     ad7e01d1d53f947c7b304cbd145b925c-2005466305.ap-south-1.elb.amazonaws.com
Port:                     <unset>  80/TCP
TargetPort:               80/TCP
NodePort:                 <unset>  31947/TCP
Endpoints:                192.168.90.198:80
Session Affinity:         None
External Traffic Policy:  Cluster
Internal Traffic Policy:  Cluster
Events:
  Type    Reason                Age    From                Message
  ----    ------                ----   ----                -------
  Normal  EnsuringLoadBalancer  2m18s  service-controller  Ensuring load balancer
  Normal  EnsuredLoadBalancer   2m16s  service-controller  Ensured load balancer
```

Checking the output using external IP: